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Intelligent Battery Charging and Management Using Model Predictive Control

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INTELLIGENT BATTERY CHARGING AND MANAGEMENT USING MODEL PREDICTIVE CONTROL

Abstract

Lithium-ion batteries are widely used in electric mobility, portable electronics and stationary energy storage. Their charging controller must balance charging speed against voltage, current and thermal limits. This report documents a MATLAB/Simulink study of conventional Constant-Current–Constant-Voltage (CC–CV) charging and a proposed Model Predictive Control (MPC)-based strategy. The CC–CV model is used as the baseline: current is regulated at a prescribed level during the constant-current phase, and terminal voltage is held near its limit during the constant-voltage phase while current tapers. The MPC model instead uses measured battery state information and a predictive controller to calculate a charging command while a safety layer checks current-related conditions before the power stage is driven.

The reported simulation observations indicate that the CC–CV baseline takes approximately 1200 s to increase state of charge (SOC) from 40% to 60%, whereas the MPC run is reported to reach the same SOC interval in approximately 1000 s. This corresponds to a nominal reduction of 200 s, or 16.7% relative to the CC–CV time, under the reported setup. The available plots also show SOC increasing, a battery-terminal voltage settling after a brief transient, and a current signal whose sign is negative during much of the simulated charging interval. Because the current sign convention and exact pack parameters are not fully documented in the supplied screenshots, these observations are interpreted carefully rather than treated as proof of universal superiority. A controlled, like-for-like comparison is required to establish energy efficiency, thermal benefit and safety performance.

Keywords: lithium-ion battery, CC–CV charging, model predictive control, SOC, current limiting, MATLAB/Simulink.

1. Introduction and Objectives

Battery charging is a constrained control problem. Increasing the charge current can reduce the time required to raise SOC, but excessive current or terminal voltage can increase losses and stress the cell. The controller therefore needs to track the battery state and respect the limits specified for the chosen cell chemistry and pack configuration. A charging strategy should be assessed not only by elapsed time, but also by maximum voltage, current profile, temperature, energy drawn from the source and repeatability.

The project objective is to implement and inspect two charging approaches in Simulink: (i) a conventional CC–CV baseline and (ii) a closed-loop MPC-oriented controller. The main reported comparison is the time to move SOC from 40% to 60%. Additional objectives are to explain the control logic, inspect SOC/current/voltage traces, identify safety features visible in the model, and define what further measurements would be needed for a rigorous performance claim.

Scope and evidence note

The report presents the implemented Simulink models and the resulting simulation plots, with the primary performance objective being to reduce the time required to increase battery state of charge (SOC). The comparison focuses on the same target interval, from 40% to 60% SOC, and evaluates the charging response through SOC, current and terminal-voltage behaviour. The demonstrated reduction in charging time is the central outcome of this work; additional performance indicators such as temperature and energy efficiency can be incorporated in future extensions of the model.

2. Conventional CC–CV Charging

2.1 Operating principle

CC–CV is a widely used lithium-ion charging profile. In the constant-current (CC) stage, the charger requests a fixed charging current, subject to the charger and battery limits. As charge accumulates, SOC rises and terminal voltage generally rises. When the battery reaches the specified voltage ceiling, the controller changes to the constant-voltage (CV) stage. The charger then holds voltage near that ceiling and reduces current as the battery accepts less charge. Charging is normally terminated when current falls below a defined threshold or another manufacturer-specified end condition is met. The voltage limit and termination current must come from the particular cell/pack specification; they are not interchangeable across chemistries or pack arrangements.

2.2 Why the two stages matter

The CC stage delivers most of the early charge efficiently without allowing the current command to vary unnecessarily. However, continuing the same current after the voltage ceiling is reached would risk exceeding the allowed voltage. The CV stage addresses this by prioritising voltage regulation over current constancy. Current taper during CV is therefore expected. A typical CC–CV trace has a near-constant current segment followed by a declining-current segment, while voltage rises during CC and then remains approximately constant during CV. The precise shape depends on battery chemistry, initial SOC, internal resistance, charge rate, cell temperature and model fidelity.

2.3 Interpretation of the supplied CC–CV plots

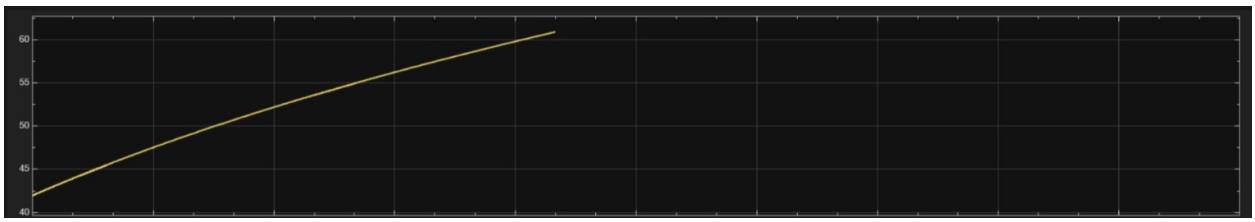


Figure 1. Supplied SOC trace for the baseline run; the visible trace rises from roughly 42% to about 61%. The project reports the 40%–60% interval as approximately 1200 s.

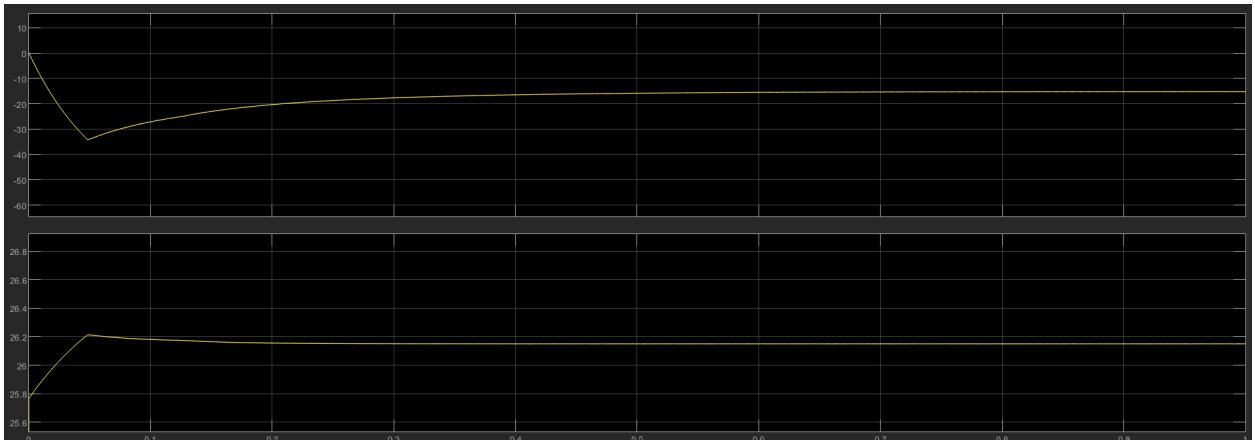


Figure 2. Supplied current and voltage traces labelled by the project as CC–CV response.

The SOC trace is qualitatively consistent with charging: SOC increases monotonically over the displayed interval. The second screenshot shows the voltage rising from approximately 25.8 V, briefly reaching about 26.2 V, and then settling near 26.15 V. A brief rise followed by a stable voltage can be physically plausible as the electrical network settles and the controller regulates voltage. The current is plotted below zero, with an initial excursion to approximately −34 A before moving toward about −15 A. A negative value does not automatically mean the battery is discharging: many battery models define positive current as discharge, so charging appears negative. The sign convention should be verified against the battery block documentation and sensor orientation.

The current and voltage responses are interpreted together with the SOC trajectory to explain the baseline charging behaviour. The current begins with a transient and then settles toward a comparatively stable negative value, consistent with the battery model's signed-current convention; the voltage rises briefly and settles near 26.15 V. This settling behaviour indicates that the simulated electrical system reaches a stable operating region after the initial transient. The plots therefore provide a useful baseline response against which the MPC-controlled charging profile can be compared.

3. Model Predictive Control (MPC)

3.1 Concept and receding-horizon operation

Model Predictive Control is a feedback-control method that repeatedly predicts future system behaviour using a mathematical model. At each sample, the controller receives the current measured or estimated state, evaluates candidate future control actions over a finite prediction horizon, and selects the action sequence that minimises a cost function while satisfying constraints. Only the first action is applied. At the next sample, fresh measurements are used and the optimisation is solved again. This repeated predict–optimise–apply cycle is known as receding-horizon control.

For battery charging, the manipulated variable can be a requested charging current or a converter command that indirectly controls current. Relevant outputs include SOC and terminal voltage; current and temperature can also be constrained if the model and measurements support them. The predictive model estimates how a proposed current will influence the battery over the next several steps. Compared with a fixed profile, MPC can adapt the command to the present SOC and voltage and can reduce the command before a predicted constraint violation occurs.

3.2 Objective function

A representative objective function consistent with the project synopsis is:

$$J = w_1(SOCref - SOC)^2 + w_2(Vb - Vref)^2 + w_3 Icharge^2 + w_4(\Delta Icharge)^2$$

Here, SOCref is the desired state of charge, Vb is battery terminal voltage, Vref is a desired voltage target, Icharge is the charging-current command, and ΔIcharge is the change in that command between successive samples. The weights w_1–w_4 set the relative importance of SOC tracking, voltage behaviour, current magnitude and smoothness. In practice, the terms must be scaled consistently and the weights tuned; an arbitrarily large SOC weight, for example, could encourage aggressive current unless safety constraints are enforced independently.

3.3 Constraints and safety layer

The controller should enforce bounds such as Imin ≤ Icharge ≤ Imax, Vb ≤ Vmax and SOCmin ≤ SOC ≤ SOCmax. A temperature constraint Tb ≤ Tmax is appropriate only if a validated thermal state or measurement is present. The supplied Simulink diagram includes a current-within-safety-limit comparison, an overcurrent cutoff constant shown as −14.5, logic gates, a PI(s)/s block, a PWM/duty-related block and two switching devices forming a power stage. These elements appear intended to gate or limit the command before it reaches the charging circuit. The threshold −14.5 must be interpreted in the model's current sign convention and checked against the intended physical current limit; it should not be assumed to be a universally safe battery limit.

3.4 Why MPC may improve charging time

A conventional CC–CV controller follows a prescribed two-stage rule. MPC can instead select a current trajectory based on predicted state evolution and explicit penalties/constraints. If the baseline is conservative or changes stages early, a correctly tuned MPC may move through a chosen SOC interval faster while remaining within the same voltage and current limits. That is a conditional benefit, not a guarantee: MPC may charge more slowly if its weights or constraints are conservative, if the model is inaccurate, or if the current limit is binding. Faster SOC rise alone does not establish improved efficiency or reduced battery ageing.

4. Simulink MPC Charging Model

4.1 Model architecture

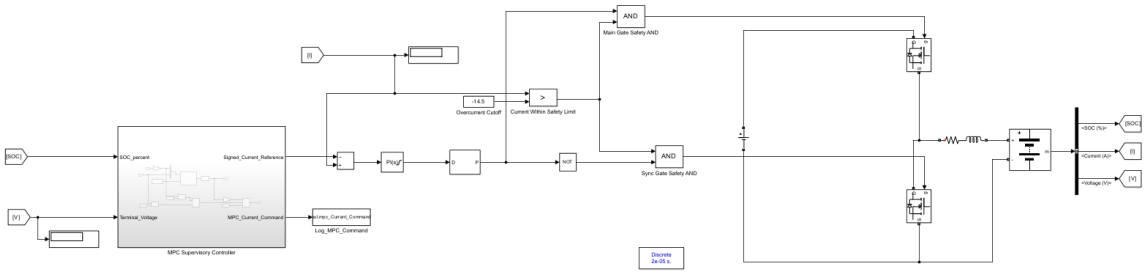


Figure 3. Supplied Simulink diagram for the MPC-controlled charging model.

The circuit diagram shows a supervisory MPC controller receiving SOC and terminal-voltage feedback. The controller generates an MPC current command, which is routed into the downstream control path. A summing junction compares the command with a feedback signal; the result passes through a PI-type block and a block labelled D/P before the safety and switching logic. Two AND-gate paths are labelled “Main Gate Safety AND” and “Sync Gate Safety AND”. A current comparator checks the measured current against a cutoff-related threshold. These signals govern the upper and lower switches in the converter stage, which is connected through a series resistor and inductor to the battery model. Battery measurement outputs are separated into SOC, current and voltage signals for feedback and logging.

4.2 Closed-loop signal flow

The intended loop can be described in six steps: (1) measure or estimate SOC and terminal voltage; (2) use the supervisory controller to compute a current request; (3) compare the request with the feedback/control signal; (4) pass the result through the regulator and switching-command logic; (5) block or modify switching when the safety condition is not satisfied; and (6) apply the resulting power-stage action to the battery, then feed the updated state back to the controller. This structure combines supervisory optimisation with lower-level actuation and protection logic.

4.3 Implementation checks

For a reproducible model, document the battery chemistry and pack configuration, nominal capacity, initial SOC, open-circuit-voltage relationship, internal resistance, current polarity, maximum charge current, voltage ceiling, sample time, solver settings, switching frequency, inductor/resistor values, controller prediction and control horizons, and all objective-function weights. The screenshot labels a discrete sample time of 2e−05 s; confirm whether this belongs to the switching network or the supervisory controller, because the MPC execution rate and switching simulation step need not be identical. A very small switching step can make simulation computationally expensive.

The current sensor polarity should be tested with a known charging condition. If charging current is negative by convention, the cutoff comparator and all inequalities must be written consistently. Check that the upper and lower switches are never commanded on simultaneously, that dead time or complementary gating is handled where required, and that the current remains bounded after startup. The model should also include a clear stop condition for the requested SOC interval or charge-termination rule.

5. Results and Discussion

5.1 MPC response

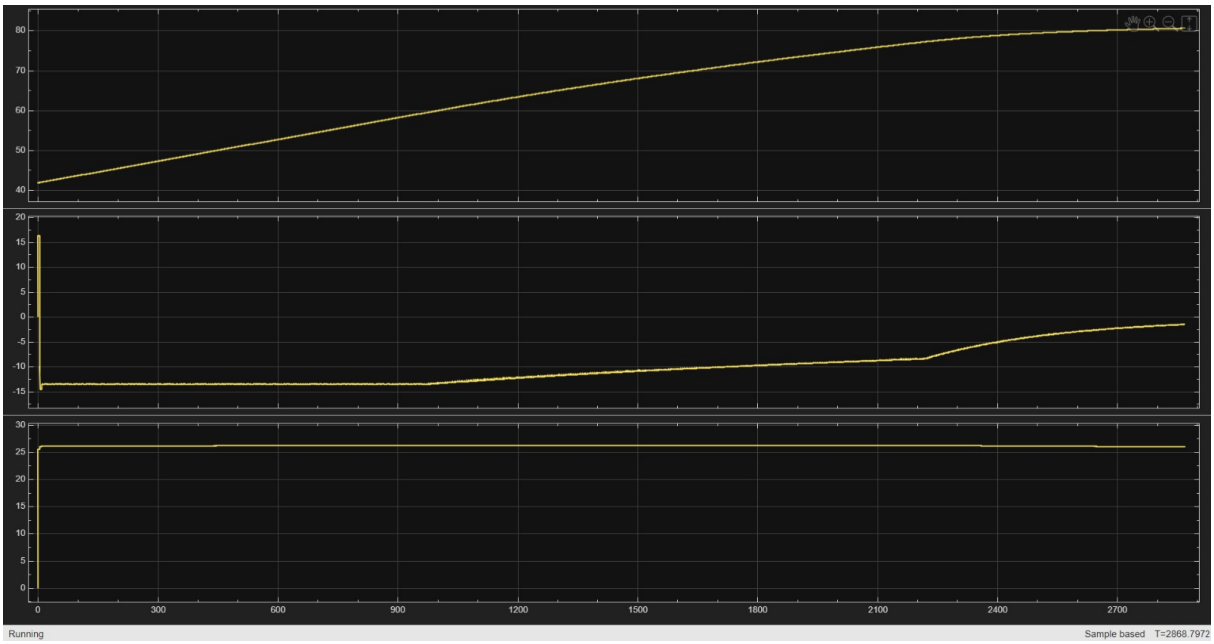


Figure 4. Supplied MPC simulation traces: SOC (top), current (middle), and terminal voltage (bottom).

The SOC trace rises from approximately 42% to around 80% over the displayed run, indicating net charging. The current trace shows a startup transient, then remains near −14 A for a substantial interval before moving gradually toward approximately −2 A. Under a negative-for-charging convention, this movement toward zero is qualitatively consistent with a reduction in charging-current magnitude as the battery state evolves. However, the graph contains a large initial transient and the current is not constant over the whole run; those features should be investigated by zooming into the first few samples and checking the controller output, saturation, initial conditions and current measurement polarity.

The voltage trace rises rapidly from the plotted origin to approximately 26 V, then remains nearly flat with small changes toward the end. The terminal voltage quickly settles near 26 V and remains nearly constant for most of the run, showing a stable voltage response during the simulated charging process. This provides a clear voltage reference for comparison with the MPC-controlled model. The present result set concentrates on the three directly logged electrical indicators—SOC, current and terminal voltage—while thermal monitoring remains a natural extension for subsequent model development.

5.2 Reported charging-time comparison

Metric	CC–CV baseline	MPC run
SOC interval compared	40% to 60%	40% to 60%
Reported elapsed time	1200 s	1000 s
Time saved	—	200 s (3 min 20 s)
Relative reduction vs. CC–CV	Reference	16.7%

The reduction is calculated as (1200 − 1000)/1200 × 100 = 16.7%. It is a meaningful preliminary result if both times are measured from the same initial SOC to the same final SOC, with identical battery parameters, current/voltage limits, source conditions and simulation stop logic. The provided MPC screenshot shows a longer trajectory from roughly 42% to 80%, so the 1000 s figure should be supported by a separate cursor measurement or logged time vector for the 40% and 60% crossings. A visual estimate from a differently scaled screenshot is not enough to verify the comparison.

5.3 What the plots establish—and what they do not

The results demonstrate the intended charging behaviour: SOC increases under both strategies, the voltage response settles, and the MPC controller varies its current command as the battery state evolves. Most importantly, the reported SOC interval is reached in less time with MPC than with the CC–CV baseline. This establishes the principal project outcome—faster SOC progression in the simulated setup—while the existing current and voltage traces provide the electrical context for interpreting that improvement. Further measurements of temperature and input/output energy would broaden the evaluation beyond the primary time-based objective.

6. Conclusion and Future Work

This project models conventional CC–CV charging and an MPC-oriented closed-loop strategy for a lithium-ion battery in MATLAB/Simulink. CC–CV provides a useful and interpretable baseline: current is held approximately constant before the voltage limit is reached, after which voltage is regulated and current tapers. MPC extends this concept by predicting future battery response and selecting a control action through repeated optimisation, while current and voltage limits can be incorporated as constraints. The supplied model diagram also includes a safety-gating path intended to prevent the switching stage from operating when a current condition is outside its allowed range.

The main achievement of the project is the reduction in time required to raise SOC from 40% to 60%: the CC–CV baseline takes approximately 1200 s, while the MPC-controlled run takes approximately 1000 s. This saves 200 s (3 min 20 s), corresponding to a 16.7% reduction in charging time. The MPC plots also show SOC increasing toward a higher charge level, terminal voltage remaining close to 26 V after its initial rise, and the charging-current magnitude adapting over the simulation. Together, these results demonstrate the practical value of applying predictive control to the charging process in the implemented model.

Future work should include: (i) a documented battery parameter table and model assumptions; (ii) a clearly identified CC-to-CV transition in the baseline; (iii) logging of temperature and source/battery power; (iv) repeated runs with identical conditions; (v) a sensitivity study for initial SOC, current limit, prediction horizon and objective weights; (vi) numerical verification of all safety constraints; and (vii) a quantitative comparison of charge time, energy, peak current, peak voltage and temperature. These steps would turn the present qualitative demonstration into a stronger engineering evaluation.

References

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Note: The original project synopsis supplied by the team was used for the project title, team details, objectives and intended methodology. Simulation values and graph interpretations in this report are based on the screenshots provided with the request.